

# Computational Mechanics

## 3. Representing the past

### Exercise sheet

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#### Instructions

Give a clear justification for each answer. You may use results proved in the lecture unless an exercise asks you to establish them. Questions marked **Extension** are optional.

### Exercise 1

#### Translating between HMM diagrams and matrices

For an edge-emitting hidden Markov model, use the convention

$$T_{ij}^{(a)} := \Pr(X_{t+1} = a, S_{t+1} = j \mid S_t = i).$$

Thus rows index the current hidden state, columns index the next hidden state, and the superscript records the emitted symbol. Throughout this exercise,  $0 < p, q < 1$ .

- (a) Suppose that  $T_{ij}^{(a)} = r$ . Explain each component of the following diagram:

$$i \xrightarrow{a:r} j.$$

What does a zero entry mean? Why is the matrix  $T := \sum_{a \in \mathcal{X}} T^{(a)}$  row-stochastic?

- (b) **From matrices to a diagram: Zero–One–Random.** Take the state order to be  $(0, 1, R)$  and suppose that

$$\eta^{(\emptyset)} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}, \quad T^{(0)} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 1-p & 0 & 0 \end{bmatrix}, \quad T^{(1)} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ p & 0 & 0 \end{bmatrix}.$$

- (i) Draw the corresponding edge-emitting HMM. Describe informally the three phases represented by 0, 1, and  $R$ .
- (ii) Calculate  $\Pr(X_{1:3} = 011)$  and  $\Pr(X_{1:3} = 000)$  in two ways: first by tracing paths through your diagram, and then from

$$\Pr(X_{1:3} = x_{1:3}) = \eta^{(\emptyset)} T^{(x_1)} T^{(x_2)} T^{(x_3)} \mathbf{1}.$$

- (c) **From a diagram to matrices: the Even Process.** Consider the edge-emitting HMM

$$A \xrightarrow{0:q} A, \quad A \xrightarrow{1:1-q} B, \quad B \xrightarrow{1:1} A,$$

with no other edges. Take the state order to be  $(A, B)$  and let  $\eta^{(\emptyset)} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix}$ .

- (i) Construct  $T^{(0)}$  and  $T^{(1)}$ . Account explicitly for every zero entry in the two matrices.
- (ii) Form  $T = T^{(0)} + T^{(1)}$  and verify that it is row-stochastic.
- (iii) Calculate  $\Pr(X_{1:3} = 011)$  and  $\Pr(X_{1:3} = 010)$  both by tracing paths through the diagram and by multiplying the appropriate symbol matrices. Explain how the second calculation reflects the defining constraint of the Even Process.
- (d) **Extension.** Let  $\mathcal{X}$  be the observation alphabet of an arbitrary finite-state HMM, let  $T = \sum_{a \in \mathcal{X}} T^{(a)}$ , and let  $\eta^{(\emptyset)}$  be a probability distribution. Prove that, for every  $n \geq 0$ ,

$$\sum_{w \in \mathcal{X}^n} \Pr(X_{1:n} = w) = \eta^{(\emptyset)} T^n \mathbf{1} = 1.$$

## Exercise 2

### Belief states and a lossy next-token readout

Continue with the Zero–One–Random HMM from Exercise 1, using state order  $(0, 1, R)$ . For an allowable history  $x_{1:n} = x_1 \cdots x_n$ , compute its belief from scratch as

$$\eta^{(x_{1:n})} := \frac{\eta^{(\emptyset)} T^{(x_1)} \cdots T^{(x_n)}}{\eta^{(\emptyset)} T^{(x_1)} \cdots T^{(x_n)} \mathbf{1}}.$$

Here  $\eta_i^{(x_{1:n})} = \Pr(S_n = i \mid X_{1:n} = x_{1:n})$ . For an allowable extension  $x_{1:n+1} = x_{1:n}a$ , update recursively as

$$\eta^{(x_{1:n+1})} = \frac{\eta^{(x_{1:n})} T^{(a)}}{\eta^{(x_{1:n})} T^{(a)} \mathbf{1}}.$$

- (a) Compute  $\eta^{(0)}$  and  $\eta^{(1)}$ . Interpret their components as posterior probabilities over the three hidden phases.
- (b) Show that the seven beliefs

$$\mathcal{B} = \{\eta^{(\emptyset)}, \eta^{(0)}, \eta^{(1)}, \eta^{(00)}, \eta^{(01)}, \eta^{(10)}, \eta^{(11)}\}$$

are distinct and that  $\mathcal{B}$  is closed under every allowable symbol update. Draw the symbol-labelled transition diagram on  $\mathcal{B}$ .

- (c) Compare your belief-state diagram with both the causal-state automaton constructed in Part 2 and the three-state generative HMM from Exercise 1.
  - (i) Match each belief to the causal state with the same shortest-length representative. Do the symbol-labelled transitions agree?
  - (ii) Which beliefs correspond to knowing the hidden phase exactly, and which represent uncertainty over several phases? Show that the hidden phase becomes known after at most three observations and remains known thereafter.
  - (iii) Observe that the finite-history belief-state presentation has seven states while the generative HMM has only three. What does this suggest about the possible relative difficulty of generation and inference?
- (d) For each belief  $\eta^{(x_{1:n})} \in \mathcal{B}$  from part (b), compute its associated next-token probability vector

$$p^{(x_{1:n})} := (\Pr(X_{n+1} = a \mid X_{1:n} = x_{1:n}))_{a \in \{0,1\}} = \eta^{(x_{1:n})} A, \quad A_{ia} := \sum_j T_{ij}^{(a)}.$$

Visualise the belief geometry on the 2-simplex over hidden states  $(0, 1, R)$  and the next-token geometry on the 1-simplex over symbols  $(0, 1)$ .

(e) Calculate

$$\Pr(X_{3:4} = 00 \mid X_{1:2} = 10) \quad \text{and} \quad \Pr(X_{3:4} = 00 \mid X_{1:2} = 01).$$

Deduce that the next-token probability vector need not be sufficient for predicting the entire future.

(f) **Extension.** Let  $\delta := \eta^{(10)} - \eta^{(01)}$ . Show that

$$\delta A = 0 \quad \text{but} \quad \delta T^{(0)} T^{(0)} \mathbf{1} \neq 0.$$

Interpret these two statements in terms of one-step and longer-horizon prediction.

### Exercise 3

#### Nonunifilar generators and large belief-state automata

For a belief  $\eta$  and a symbol  $a$  satisfying  $\eta T^{(a)} \mathbf{1} > 0$ , write

$$F_a(\eta) := \frac{\eta T^{(a)}}{\eta T^{(a)} \mathbf{1}}.$$

Thus the belief-state automaton contains the transition

$$\eta \xrightarrow{a: \eta T^{(a)} \mathbf{1}} F_a(\eta).$$

(a) **The Simple Nonunifilar Source.** Revisit the source from Part 2. With state order  $(A, B)$ , its symbol matrices and initial belief are

$$T^{(0)} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} \end{bmatrix}, \quad T^{(1)} = \begin{bmatrix} 0 & 0 \\ \frac{1}{2} & 0 \end{bmatrix}, \quad \eta^{(\emptyset)} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

(i) For  $n \geq 0$ , show that

$$\eta^{(10^n)} = \begin{bmatrix} \frac{1}{n+1} & \frac{n}{n+1} \end{bmatrix}, \quad \eta^{(0^n)} = \begin{bmatrix} \frac{1}{n+2} & \frac{n+1}{n+2} \end{bmatrix}.$$

Deduce that  $\eta^{(0^n)} = \eta^{(10^{n+1})}$  and observe that  $\eta^{(\emptyset)} = \eta^{(10)}$ .

(ii) Show that, for every  $n \geq 1$ ,

$$\eta^{(10^n 1)} = \eta^{(1)}.$$

(iii) Draw the first four states of the belief-state automaton and indicate its continuing structure with an ellipsis. Show that the beliefs  $\eta^{(10^n)}$  are all distinct, identify their limiting point in the 1-simplex, and explain why the two-state generator produces a countably infinite belief-state automaton.

(b) **Mess3.** Now take hidden-state and symbol sets both equal to  $\{0, 1, 2\}$ , with

$$\alpha = \frac{1}{5}, \quad b = \frac{1-\alpha}{2} = \frac{2}{5}, \quad x = \frac{3}{20}, \quad y = 1 - 2x = \frac{7}{10},$$

and

$$T^{(0)} = \begin{bmatrix} \alpha y & bx & bx \\ \alpha x & by & bx \\ \alpha x & bx & by \end{bmatrix}, \quad T^{(1)} = \begin{bmatrix} by & \alpha x & bx \\ bx & \alpha y & bx \\ bx & \alpha x & by \end{bmatrix},$$

$$T^{(2)} = \begin{bmatrix} by & bx & \alpha x \\ bx & by & \alpha x \\ bx & bx & \alpha y \end{bmatrix}, \quad \eta^{(\emptyset)} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}.$$

- (i) Verify that  $T = \sum_{a=0}^2 T^{(a)}$  is row-stochastic and explain why this generator is nonunifilar. Compute  $\eta^{(0)}$ ,  $\eta^{(1)}$ , and  $\eta^{(2)}$ .
- (ii) Compute  $\eta^{(00)}$ ,  $\eta^{(01)}$ , and  $\eta^{(02)}$ . Use the cyclic symmetry of the process to sketch the first two generations of the belief-state automaton, including the symbol labels.
- (iii) Compute the next-token matrix  $A$ , with  $A_{ia} = \sum_j T_{ij}^{(a)}$ , and show that it is invertible. What does this imply about whether distinct beliefs can be merged by the map  $p^{(x_{1:n})} = \eta^{(x_{1:n})} A$ ? Contrast this with Exercise 2.
- (iv) Using a short program, form

$$\mathcal{B}_d := \{\eta^{(x_{1:d})} : x_{1:d} \in \{0, 1, 2\}^d\}, \quad 0 \leq d \leq 7.$$

Record the number of distinct beliefs at each depth. Visualise  $\mathcal{B}_7$  on the hidden-state 2-simplex, colouring each point by its final symbol. Also visualise the associated next-token vectors on the symbol 2-simplex. Describe the geometric structure that emerges and how the three maps  $F_0, F_1, F_2$  generate the outgoing transitions of the belief-state automaton.

- (v) Explain why the belief-state automata constructed in this exercise are unifilar even though their generators are not. Compare the ladder-with-resets structure of the Simple Nonunifilar Source with the branching structure of Mess3. Finally, explain why the set of beliefs reached by finite histories is countable even though its closure can have a much richer fractal geometry.

## Exercise 4

### Generalised hidden Markov models and predictive vectors

A  $d$ -dimensional generalised hidden Markov model (GHMM) consists of a finite alphabet  $\mathcal{X}$ , real  $d \times d$  matrices  $(T^{(a)})_{a \in \mathcal{X}}$ , an initial row vector  $\eta^{(\emptyset)}$ , and a column vector  $\phi$ . Writing

$$T := \sum_{a \in \mathcal{X}} T^{(a)},$$

these objects satisfy

$$T\phi = \phi, \quad \eta^{(\emptyset)}\phi = 1, \quad \eta^{(\emptyset)}T^{(x_1)} \dots T^{(x_n)}\phi \geq 0$$

for every  $x_{1:n} \in \mathcal{X}^n$  and every  $n \geq 0$ .

- (a) **A probability law from matrix products.** Consider

$$\Pr(X_{1:n} = x_{1:n}) = \eta^{(\emptyset)}T^{(x_1)} \dots T^{(x_n)}\phi.$$

- (i) Show that these quantities are nonnegative and that

$$\sum_{x_{1:n} \in \mathcal{X}^n} \Pr(X_{1:n} = x_{1:n}) = 1$$

for every  $n \geq 0$ .

- (ii) Show that

$$\sum_{a \in \mathcal{X}} \Pr(X_{1:n+1} = x_{1:n}a) = \Pr(X_{1:n} = x_{1:n}).$$

Explain why this is the consistency condition needed to define a one-sided stochastic process.

- (iii) The GHMM conditions do not require  $\eta^{(\emptyset)}T = \eta^{(\emptyset)}$ . Explain why the process need not therefore be stationary. Show that this additional equality is sufficient for stationarity.

- (b) **Predictive vectors.** For any  $x_{1:n}$  satisfying  $\Pr(X_{1:n} = x_{1:n}) > 0$ , define its predictive vector by

$$\eta^{(x_{1:n})} := \frac{\eta^{(\emptyset)} T^{(x_1)} \dots T^{(x_n)}}{\eta^{(\emptyset)} T^{(x_1)} \dots T^{(x_n)} \phi}.$$

- (i) Show that

$$\eta^{(x_{1:n})} \phi = 1$$

and, for every continuation  $y_{1:m}$ ,

$$\Pr(X_{n+1:n+m} = y_{1:m} \mid X_{1:n} = x_{1:n}) = \eta^{(x_{1:n})} T^{(y_1)} \dots T^{(y_m)} \phi.$$

- (ii) Derive the recursive update and identify its normalising factor:

$$\eta^{(x_{1:n}a)} = \frac{\eta^{(x_{1:n})} T^{(a)}}{\eta^{(x_{1:n})} T^{(a)} \phi}, \quad \Pr(X_{n+1} = a \mid X_{1:n} = x_{1:n}) = \eta^{(x_{1:n})} T^{(a)} \phi.$$

- (iii) Conceptually contrast an HMM belief state with a GHMM predictive vector. Compare the meaning of their coordinates, their positivity and normalisation properties, and how they encode predictions of future observations.

- (c) **Predictive vectors need not be probability vectors.** Consider the two-state HMM

$$M^{(0)} = \begin{bmatrix} \frac{4}{5} & 0 \\ 0 & \frac{1}{5} \end{bmatrix}, \quad M^{(1)} = \begin{bmatrix} \frac{1}{5} & 0 \\ 0 & \frac{4}{5} \end{bmatrix}, \quad \alpha = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

Thus a hidden state is selected initially and thereafter emits a biased coin without changing state. Let

$$S = \begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix}, \quad T^{(a)} := S^{-1} M^{(a)} S, \quad \eta^{(\emptyset)} := \alpha S, \quad \phi := S^{-1} \mathbf{1}.$$

- (i) Compute  $T^{(0)}, T^{(1)}, \eta^{(\emptyset)}$ , and  $\phi$ . Show that, for every  $x_{1:n}$ ,

$$\eta^{(\emptyset)} T^{(x_1)} \dots T^{(x_n)} \phi = \alpha M^{(x_1)} \dots M^{(x_n)} \mathbf{1}.$$

Deduce that this change of coordinates gives a valid GHMM for the same process.

- (ii) Compute  $\eta^{(0)}$  and  $\eta^{(1)}$ . Verify directly from  $\eta^{(0)}$  that

$$\Pr(X_2 = 0 \mid X_1 = 0) = \frac{17}{25}, \quad \Pr(X_2 = 1 \mid X_1 = 0) = \frac{8}{25}.$$

Explain why the entries of a GHMM predictive vector need not themselves be probabilities.

- (d) **The probability matrix and minimal dimension.** For finite sets  $\mathcal{U}, \mathcal{V} \subseteq \mathcal{X}^*$ , define the probability matrix by

$$P_{\mathcal{U}, \mathcal{V}}(u, v) := \Pr(X_{1:|u|+|v|} = uv), \quad u \in \mathcal{U}, \quad v \in \mathcal{V}.$$

- (i) Use the GHMM matrix product to factor  $P_{\mathcal{U}, \mathcal{V}}$  as a matrix of prefix row vectors times a matrix of continuation column vectors. Deduce that

$$\text{rank } P_{\mathcal{U}, \mathcal{V}} \leq d$$

for every  $d$ -dimensional GHMM realisation.

- (ii) For the process in part (c), take  $\mathcal{U} = \mathcal{V} = \{\emptyset, 0, 1\}$ . Compute  $P_{\mathcal{U}, \mathcal{V}}$  and use an SVD to find its singular values. Deduce that the two-dimensional GHMM is minimal-dimensional.